

CLASS 8 CHEMISTRY FOUNDATION

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Formula Sheets
Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 1: Synthetic Fibres and Plastics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that cotton, wool and silk are natural fibres from plants and animals.
- Basic understanding that some materials burn and others melt.
- Familiarity with everyday objects made of plastic.
- Knowledge that materials can be classified by their properties.

1.1 Synthetic Fibres — Rayon, Nylon, Polyester and Acrylic

 **Estimated time:** 15 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Synthetic fibres are man-made fibres produced by joining small chemical units called monomers into long chains called polymers, and the four common ones are rayon, nylon, polyester and acrylic.

Real-Life Example

A nylon parachute, a polyester school shirt, an acrylic sweater and a rayon saree are four different synthetic fibres you can find in one wardrobe.

Detailed Explanation

Definition

A polymer is a very large molecule formed by joining many small repeating units called monomers. A synthetic fibre is a fibre made from such polymers by chemical processes rather than obtained from a plant or animal.

Key Idea

Rayon is the odd one out — it is made from wood pulp (natural cellulose) that has been chemically treated, so it is called a semi-synthetic or regenerated fibre, and is known as artificial silk. All the others are made entirely from petrochemicals.

Working Principle

Nylon was the first fully synthetic fibre, made in 1931 from coal, water and air. It is exceptionally strong, elastic and lustrous. Polyester (terylene, PET) resists wrinkles and is easy to wash. Acrylic imitates wool and is warm, light and cheap.

Applications

- Nylon: ropes, parachutes, seat belts, tents, toothbrush bristles, fishing nets.
- Polyester: terrycot and terrywool blends, PET bottles, sails.
- Acrylic: sweaters, shawls, blankets — 'artificial wool'.

✓ Points to Remember (Exam Focus)

- ✓ Monomer → polymer; polymerisation is the joining reaction.
- ✓ Rayon is semi-synthetic, made from wood pulp — called artificial silk.
- ✓ Nylon is the strongest common synthetic fibre; a nylon thread is stronger than a steel wire of the same thickness.
- ✓ Polyester is made of repeating ester units and does not wrinkle easily.
- ✓ Acrylic is used as an inexpensive substitute for wool.

⚠ Common Student Mistakes

- ⚠ Calling rayon fully synthetic — it is regenerated from natural cellulose.
- ⚠ Saying synthetic fibres absorb water like cotton; they absorb very little, which is why they dry fast but feel uncomfortable in summer.
- ⚠ Wearing synthetic clothes while working in the kitchen or a laboratory — they melt and stick to the skin.

**Memory Hook**

'RaNPA' — Rayon, Nylon, Polyester, Acrylic; and 'Rayon is the Real (natural) one among the fakes'.

**Advanced Insights**

Synthetic fibres are thermoplastic: on heating they soften and melt rather than char like cotton. This is exactly why fire-safety rules forbid synthetics near flames — molten polymer sticks to skin and causes far deeper burns than a cotton fire would.

1.2 Plastics — Types, Properties and Uses

⌚ Estimated time: 14 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Plastics are polymers that can be moulded into any shape, and they are of two kinds — thermoplastics, which soften on heating and can be remoulded, and thermosetting plastics, which set permanently.

**Real-Life Example**

A polythene bag can be melted and re-made, but a burnt electrical switch or a cracked melamine plate can never be reshaped — the first is a thermoplastic, the second thermosetting.

Detailed Explanation

Definition

Plastic is a polymer that can be moulded into any shape under heat and pressure. Thermoplastics soften on heating and can be remoulded repeatedly; thermosetting plastics, once set, cannot be softened again.

Key Idea

The difference lies in the arrangement of the chains. Thermoplastics have linear or slightly branched chains that slide over one another when heated; thermosetting plastics have chains cross-linked into a rigid three-dimensional network.

Working Principle

Plastics are chosen because they are light, strong, durable, non-reactive, poor conductors of heat and electricity, and cheap to produce. These same properties — especially non-reactivity — make them very difficult to dispose of.

Applications

- Thermoplastics: polythene (bags), PVC (pipes), polystyrene (packing), polypropylene (containers).
- Thermosetting plastics: bakelite (electrical switches, handles), melamine (crocery, floor tiles).
- Teflon is used for non-stick cookware; PVC for raincoats and insulation.

✓ Points to Remember (Exam Focus)

- ✓ Thermoplastic = can be softened and remoulded; e.g. polythene, PVC.
- ✓ Thermosetting = sets permanently, cannot be remoulded; e.g. bakelite, melamine.
- ✓ Bakelite is a poor conductor of heat and electricity — used for switches and cooker handles.
- ✓ Melamine resists fire and is used for kitchenware and fire-fighting uniforms.
- ✓ Plastics are non-reactive, light, durable and cheap — the reason they are everywhere.

⚠ Common Student Mistakes

- ⚠ Thinking all plastics can be melted and recycled — thermosetting plastics cannot.
- ⚠ Confusing bakelite (thermosetting) with polythene (thermoplastic).
- ⚠ Saying plastics are strong because they are heavy; they are strong and light, which is the point.



Memory Hook

'ThermoPLASTIC = PLIABLE again and again; ThermoSET = SET for good.'



Advanced Insights

The polymer chains in a thermoset are joined by covalent cross-links, so melting would require breaking chemical bonds rather than just overcoming weak forces between chains. That is why a thermoset chars and decomposes instead of melting.

II Quick Recap — Topics Covered So Far

- **Synthetic Fibres — Rayon, Nylon, Polyester and Acrylic** — Synthetic fibres are man-made fibres produced by joining small chemical units called monomers into long chains called polymers, and the four common ones are rayon, nylon, polyester and acrylic.
- **Plastics — Types, Properties and Uses** — Plastics are polymers that can be moulded into any shape, and they are of two kinds — thermoplastics, which soften on heating and can be remoulded, and thermosetting plastics, which set permanently.

1.3 Plastics and the Environment

🕒 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Plastics are non-biodegradable, so they persist for hundreds of years, choke drains and animals, and release poisonous gases when burnt — which is why the 4 R's matter.

Real-Life Example

The plastic bags that block city drains during the monsoon, and the plastic found in the stomachs of cows and sea turtles, are the direct cost of convenience.

Detailed Explanation

Definition

A biodegradable material is one that is decomposed by natural processes such as the action of bacteria; a non-biodegradable material is one that is not, or that takes an extremely long time.

Key Idea

The very property that makes plastic useful — chemical non-reactivity — is the property that makes it an environmental problem. Microorganisms have no enzymes that can break the long carbon chains.

Working Principle

Burning plastic does not solve the problem: it releases toxic gases such as dioxins and carbon monoxide. The workable answers are to Reduce use, Reuse items, Recycle thermoplastics, and Recover energy where recycling is impossible.

Applications

- Carrying a cloth or jute bag instead of accepting plastic bags.
- Segregating waste into biodegradable and non-biodegradable at source.
- Recycling PET bottles into polyester fibre for fleece and filling material.

✓ **Points to Remember (Exam Focus)**

- ✓ Plastics are non-biodegradable and may persist for hundreds of years.
- ✓ Burning plastic releases poisonous gases and must be avoided.
- ✓ The 4 R's: Reduce, Reuse, Recycle, Recover.
- ✓ Only thermoplastics can be recycled; thermosetting plastics cannot.
- ✓ Plastic waste is a major cause of blocked drains, soil damage and animal deaths.

⚠ Common Student Mistakes

- ⚠ Believing plastic 'disappears' if buried — it only breaks into microplastics.
- ⚠ Assuming all plastic waste is recyclable; contaminated and thermosetting plastic is not.
- ⚠ Thinking incineration is a clean disposal method for plastics.



Memory Hook

'Reduce first, Recycle last' — and 'What does not rot, must not be bought in bulk'.



Advanced Insights

Microplastics — fragments below 5 mm — now reach the human body through drinking water, salt and seafood. Biodegradable plastics made from polylactic acid (from maize starch) exist, but they need industrial composting conditions and do not break down in an ordinary landfill.

II Quick Recap — Topics Covered So Far

- Plastics and the Environment — Plastics are non-biodegradable, so they persist for hundreds of years, choke drains and animals, and release poisonous gases when burnt — which is why the 4 R's matter.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Synthetic Fibres — Rayon, Nylon, Polyester and Acrylic: 'RaNPA' — Rayon, Nylon, Polyester, Acrylic; and 'Rayon is the Real (natural) one among the fakes'.
- Plastics — Types, Properties and Uses: 'ThermoPLASTIC = PLIABLE again and again; ThermoSET = SET for good.'
- Plastics and the Environment: 'Reduce first, Recycle last' — and 'What does not rot, must not be bought in bulk'.

All Common Mistakes

-  Calling rayon fully synthetic — it is regenerated from natural cellulose.
-  Saying synthetic fibres absorb water like cotton; they absorb very little, which is why they dry fast but feel uncomfortable in summer.
-  Wearing synthetic clothes while working in the kitchen or a laboratory — they melt and stick to the skin.
-  Thinking all plastics can be melted and recycled — thermosetting plastics cannot.
-  Confusing bakelite (thermosetting) with polythene (thermoplastic).
-  Saying plastics are strong because they are heavy; they are strong and light, which is the point.
-  Believing plastic 'disappears' if buried — it only breaks into microplastics.
-  Assuming all plastic waste is recyclable; contaminated and thermosetting plastic is not.
-  Thinking incineration is a clean disposal method for plastics.

Quick Revision Section

- Synthetic Fibres — Rayon, Nylon, Polyester and Acrylic: Synthetic fibres are man-made fibres produced by joining small chemical units called monomers into long chains called polymers, and the four common ones are rayon, nylon, polyester and acrylic.
- Plastics — Types, Properties and Uses: Plastics are polymers that can be moulded into any shape, and they are of two kinds — thermoplastics, which soften on heating and can be remoulded, and thermosetting plastics, which set permanently.
- Plastics and the Environment: Plastics are non-biodegradable, so they persist for hundreds of years, choke drains and animals, and release poisonous gases when burnt — which is why the 4 R's matter.

Chapter 2: Materials — Metals and Non-metals

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that materials differ in hardness, shine and conductivity.
- Familiarity with common metals such as iron, copper and aluminium.
- Basic understanding of a chemical reaction.
- Knowledge that some substances burn in air.

2.1 Physical Properties of Metals and Non-metals

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Metals are lustrous, malleable, ductile, sonorous and good conductors of heat and electricity, while non-metals are generally dull, brittle and poor conductors.

Real-Life Example

A copper wire carries current and can be drawn thin, but a piece of coal crumbles to powder when hammered — one is a metal, the other a non-metal.

Detailed Explanation

Definition

Malleability is the property of being beaten into thin sheets; ductility is the property of being drawn into wires; sonorousness is the property of producing a ringing sound when struck.

Key Idea

These properties follow from metallic bonding — positive metal ions in a 'sea' of free electrons. The free electrons carry heat and electricity, and let layers of ions slide over one another without the metal breaking, which is why metals bend instead of shattering.

Working Principle

Non-metals have no free electrons, so they neither conduct nor deform; when hammered they simply break. Every general rule has exceptions, and those exceptions are a favourite examination question.

Applications

- Copper and aluminium are used for electrical wiring because of high conductivity.
- Gold and silver are the most malleable and ductile metals, used for jewellery and gold leaf.
- Graphite, a non-metal, conducts electricity and is used for electrodes and pencil leads.

✓ Points to Remember (Exam Focus)

- ✓ Metals: lustrous, malleable, ductile, sonorous, good conductors, high density and melting point.
- ✓ Non-metals: dull, brittle, non-sonorous, poor conductors, low melting point.
- ✓ Exceptions among metals: mercury is liquid; sodium and potassium are soft and cut with a knife; lead is a poor conductor.
- ✓ Exceptions among non-metals: graphite conducts electricity; iodine is lustrous; diamond is the hardest known substance.
- ✓ Metalloids such as silicon, boron and arsenic show properties of both.

⚠ Common Student Mistakes

- ⚠ Saying all metals are solid at room temperature — mercury is a liquid.
- ⚠ Claiming no non-metal conducts electricity; graphite does.
- ⚠ Confusing malleability (sheets) with ductility (wires).

**Memory Hook**

'Malleable = Mallet makes sheets; Ductile = Drawn into wires' — and 'Mercury is the Metal that Moves'.

**Advanced Insights**

Diamond and graphite are both pure carbon, yet diamond is the hardest insulator and graphite a soft conductor. The difference is entirely structural: diamond bonds all four valence electrons in a rigid 3D lattice, while graphite leaves one electron free in flat sliding sheets.

2.2 Chemical Properties and the Reactivity Series

Estimated time: 16 mins **Difficulty:** Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Metals react with oxygen to form basic oxides and with acids to release hydrogen, while non-metals form acidic oxides; the reactivity series ranks metals by how vigorously they react.

**Real-Life Example**

Sodium is stored under kerosene because it catches fire in air, whereas gold rings stay bright for generations — the two ends of the reactivity series.

**Detailed Explanation**

Definition

The reactivity series is a list of metals arranged in decreasing order of their chemical reactivity. A displacement reaction is one in which a more reactive metal displaces a less reactive metal from its salt solution.

Key Idea

Metal oxides are basic and turn red litmus blue when dissolved; non-metal oxides are acidic and turn blue litmus red. Some oxides, such as aluminium oxide and zinc oxide, are amphoteric — they react with both acids and bases.

Working Principle

A metal above another in the series will displace it from solution. Iron displaces copper from copper sulphate solution (blue solution turns green, red-brown copper deposits), but copper cannot displace iron.

Applications

- Metals above hydrogen in the series release hydrogen gas with dilute acids; copper, silver and gold do not.
- Iron nails placed in copper sulphate solution get coated with copper.
- Thermite reaction: aluminium displaces iron from iron oxide, used to weld railway tracks.

✓ Points to Remember (Exam Focus)

- ✓ Reactivity series: $K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au$.
- ✓ Metal + oxygen $\rightarrow$ basic oxide; Non-metal + oxygen $\rightarrow$ acidic oxide.
- ✓ Metal + dilute acid $\rightarrow$ salt + hydrogen gas (burns with a pop sound).
- ✓ Copper, silver and gold do NOT react with dilute acids to give hydrogen.
- ✓ Aluminium and zinc oxides are amphoteric — they behave as both acid and base.

⚠ Common Student Mistakes

- ⚠ Expecting copper to displace hydrogen from acid — copper lies below hydrogen in the series.
- ⚠ Saying all metal oxides are basic; aluminium and zinc oxides are amphoteric.
- ⚠ Reversing displacement — the MORE reactive metal displaces the less reactive one.

**Memory Hook**

'Please Send Cats, Monkeys And Zebras In Lead Cages Made Securely Gold-plated' — K, Na, Ca, Mg, Al, Zn, Fe, Pb, Cu, Hg, Ag, Au.

**Advanced Insights**

The reactivity series is really an ordering by how readily a metal loses electrons, measured as standard electrode potential. This is why the same series predicts both displacement reactions and which metal will corrode first in a galvanic pair.

II Quick Recap — Topics Covered So Far

- Physical Properties of Metals and Non-metals — Metals are lustrous, malleable, ductile, sonorous and good conductors of heat and electricity, while non-metals are generally dull, brittle and poor conductors.
- Chemical Properties and the Reactivity Series — Metals react with oxygen to form basic oxides and with acids to release hydrogen, while non-metals form acidic oxides; the reactivity series ranks metals by how vigorously they react.

2.3 Corrosion, Extraction of Metals and Alloys

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Corrosion is the slow eating away of a metal by air and moisture; metals are extracted from ores by concentration, reduction and refining, and alloys are mixtures made to improve properties.

Real-Life Example

A rusted iron gate, a green patina on an old brass idol, and stainless-steel kitchen utensils that never rust are corrosion and its cure side by side.

Detailed Explanation

Definition

Corrosion is the gradual destruction of a metal by chemical reaction with air, moisture or chemicals. An alloy is a homogeneous mixture of a metal with one or more metals or non-metals.

Key Idea

Rusting of iron needs BOTH air (oxygen) and moisture — remove either one and rusting stops. Rust is hydrated iron(III) oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, and it flakes off exposing fresh metal, so iron corrodes right through.

Working Principle

Metallurgy has three stages: concentration of the ore (removing gangue), reduction of the concentrated ore to free metal, and refining of the impure metal. Highly reactive metals are extracted by electrolysis, moderately reactive ones by reduction with carbon.

Applications

- Prevention of rusting: painting, oiling, greasing, galvanisation (zinc coating), electroplating, alloying.
- Alloys: steel (Fe + C), stainless steel (Fe + Cr + Ni), brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn).
- Aluminium is extracted from bauxite by electrolysis because it is highly reactive.

✓ **Points to Remember (Exam Focus)**

- ✓ Rusting requires both oxygen and water; rust is $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, reddish-brown and flaky.
- ✓ Galvanisation is coating iron with a layer of zinc, which protects even if scratched.
- ✓ Ore = mineral from which a metal can be profitably extracted; gangue = unwanted impurities.
- ✓ Alloys are harder, stronger and more corrosion-resistant than pure metals.
- ✓ Amalgam is any alloy in which mercury is one component.

⚠ Common Student Mistakes

- ⚠ Thinking rusting happens with air alone — dry air causes no rusting.
- ⚠ Calling every mineral an ore; only those from which extraction is economical are ores.
- ⚠ Saying alloys are compounds — they are homogeneous mixtures, not compounds.

🧠 Memory Hook

'Rust needs Air + Water — remove one, rust is gone' — and 'BRASS = copper + zinc, BRONZE = copper + tin'.

🚀 Advanced Insights

Galvanisation protects by sacrificial corrosion: zinc is more reactive than iron, so even at a scratch the zinc corrodes preferentially and the exposed iron is left untouched. A tin coating, by contrast, accelerates rusting once scratched, because tin is less reactive than iron.

II Quick Recap — Topics Covered So Far

- Corrosion, Extraction of Metals and Alloys — Corrosion is the slow eating away of a metal by air and moisture; metals are extracted from ores by concentration, reduction and refining, and alloys are mixtures made to improve properties.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Physical Properties of Metals and Non-metals: 'Malleable = Mallet makes sheets; Ductile = Drawn into wires' — and 'Mercury is the Metal that Moves'.
- Chemical Properties and the Reactivity Series: 'Please Send Cats, Monkeys And Zebras In Lead Cages Made Securely Gold-plated' — K, Na, Ca, Mg, Al, Zn, Fe, Pb, Cu, Hg, Ag, Au.
- Corrosion, Extraction of Metals and Alloys: 'Rust needs Air + Water — remove one, rust is gone' — and 'BRASS = coppeR + zinc, BRONZE = coppeR + tiN'.

All Common Mistakes

-  Saying all metals are solid at room temperature — mercury is a liquid.
-  Claiming no non-metal conducts electricity; graphite does.
-  Confusing malleability (sheets) with ductility (wires).
-  Expecting copper to displace hydrogen from acid — copper lies below hydrogen in the series.
-  Saying all metal oxides are basic; aluminium and zinc oxides are amphoteric.
-  Reversing displacement — the MORE reactive metal displaces the less reactive one.
-  Thinking rusting happens with air alone — dry air causes no rusting.
-  Calling every mineral an ore; only those from which extraction is economical are ores.
-  Saying alloys are compounds — they are homogeneous mixtures, not compounds.

Quick Revision Section

- Physical Properties of Metals and Non-metals: Metals are lustrous, malleable, ductile, sonorous and good conductors of heat and electricity, while non-metals are generally dull, brittle and poor conductors.
- Chemical Properties and the Reactivity Series: Metals react with oxygen to form basic oxides and with acids to release hydrogen, while non-metals form acidic oxides; the reactivity series ranks metals by how vigorously they react.
- Corrosion, Extraction of Metals and Alloys: Corrosion is the slow eating away of a metal by air and moisture; metals are extracted from ores by concentration, reduction and refining, and alloys are mixtures made to improve properties.

Chapter 3: Coal and Petroleum

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that fuels burn to give heat and light.
- Knowledge that some resources are limited on earth.
- Basic difference between solid, liquid and gaseous substances.
- Familiarity with LPG cylinders and petrol pumps.

3.1 Natural Resources and the Formation of Fossil Fuels

 **Estimated time:** 13 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Coal, petroleum and natural gas are fossil fuels formed over millions of years from buried remains of ancient plants and animals, and they are exhaustible natural resources.

Real-Life Example

The petrol in a scooter today was living plankton in a sea three hundred million years ago — which is why it can never be replaced within a human lifetime.

Detailed Explanation

Definition

Inexhaustible natural resources are present in unlimited quantity and are not likely to be exhausted (air, sunlight). Exhaustible natural resources are limited in quantity and can be used up (coal, petroleum, natural gas, minerals, forests).

Key Idea

Fossil fuels form only under very specific conditions — burial without oxygen, plus high temperature and pressure, sustained for millions of years. That is why they are non-renewable on any practical timescale.

Working Principle

Coal formed when dense forests were buried by earth movements about 300 million years ago; the slow conversion of dead vegetation into coal is called carbonisation. Petroleum and natural gas formed from the remains of marine organisms buried under sand and clay.

Applications

- Petroleum is found trapped between two layers of impervious rock, with natural gas above it.
- Coal is mined; petroleum is drilled and pumped from oil wells.
- The stages of coal formation: peat → lignite → bituminous → anthracite, in increasing carbon content.

✓ Points to Remember (Exam Focus)

- ✓ Fossil fuels: coal, petroleum, natural gas — all exhaustible and non-renewable.
- ✓ Carbonisation is the slow conversion of dead vegetation into coal.
- ✓ Petroleum is lighter than water and floats above it in the rock strata.
- ✓ Anthracite has the highest carbon content and is the best quality coal.
- ✓ Fossil fuels took millions of years to form and cannot be replaced in our lifetime.

⚠ Common Student Mistakes

- ⚠ Calling forests inexhaustible — they are exhaustible, though renewable if managed.
- ⚠ Saying coal is formed from dead animals; coal comes from plants, petroleum from marine organisms.
- ⚠ Confusing carbonisation (coal formation) with combustion (burning).

**Memory Hook**

'Coal from Trees, Petroleum from the Seas' — and coal grades 'Peat, Lignite, Bituminous, Anthracite' get better as you go.

**Advanced Insights**

Burning fossil fuels releases carbon that was locked away for hundreds of millions of years, in a couple of centuries. The rate mismatch — slow burial, fast release — is the fundamental reason atmospheric CO₂ is rising so sharply.

3.2 Coal and Its Products

🕒 **Estimated time:** 13 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Heating coal in the absence of air is called destructive distillation, and it yields coke, coal tar and coal gas — three products with distinct uses.

**Real-Life Example**

The naphthalene balls in a cupboard, the black tar on a road, and the coke used in a steel plant all begin as the same lump of coal.

**Detailed Explanation****Definition**

Destructive distillation is the process of heating coal strongly in the absence of air. Coke is a tough, porous, black substance that is almost pure carbon.

Key Idea

Destructive distillation must exclude air — if air were present, the coal would simply burn to ash and carbon dioxide instead of decomposing into useful products.

Working Principle

On strong heating without air, coal decomposes into a solid residue (coke), a thick black liquid (coal tar), and a mixture of gases (coal gas). Coal tar is a mixture of about 200 substances and was once the main source of dyes and drugs.

Applications

- Coke: used as a reducing agent in the extraction of metals and in the manufacture of steel.
- Coal tar: source of naphthalene balls, paints, explosives, synthetic dyes, plastics, roofing material.
- Coal gas: used as a fuel; it was used for street lighting in London from 1810 and in India from 1897.

✓ Points to Remember (Exam Focus)

- ✓ Destructive distillation = strong heating in the ABSENCE of air.
- ✓ Products of coal: coke (solid), coal tar (liquid), coal gas (gas).
- ✓ Coke is almost pure carbon and is used as a reducing agent in metallurgy.
- ✓ Bitumen, a petroleum product, has now replaced coal tar for surfacing roads.
- ✓ Charcoal is obtained by heating wood in limited air, not coal.

⚠ Common Student Mistakes

- ⚠ Saying destructive distillation happens in the presence of air — the absence of air is the whole point.
- ⚠ Confusing coke (from coal) with charcoal (from wood).
- ⚠ Thinking coal gas is the same as natural gas; coal gas is manufactured, natural gas is extracted.



Memory Hook

'Coal gives Coke, Coal Tar and Coal Gas — one Solid, one Liquid, one Gas.'



Advanced Insights

Coal tar was the birthplace of the synthetic organic chemical industry. Perkin's accidental discovery of mauveine in 1856 while working with a coal-tar derivative created the first synthetic dye and launched an entire industry from what had been an industrial waste product.

II Quick Recap — Topics Covered So Far

- Natural Resources and the Formation of Fossil Fuels — Coal, petroleum and natural gas are fossil fuels formed over millions of years from buried remains of ancient plants and animals, and they are exhaustible natural resources.
- Coal and Its Products — Heating coal in the absence of air is called destructive distillation, and it yields coke, coal tar and coal gas — three products with distinct uses.

3.3 Petroleum, Natural Gas and Conservation

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Petroleum is separated into useful fractions by fractional distillation in a refinery, natural gas is a clean fuel stored as CNG, and both must be conserved because they are running out.

Real-Life Example

The same crude oil yields the LPG in your kitchen, the petrol in a bike, the diesel in a bus and the tar on the road — sorted purely by boiling point.

Detailed Explanation

Definition

Petroleum is a dark, oily liquid mixture of many hydrocarbons, also called crude oil or black gold. Refining is the process of separating crude oil into its useful constituents, called fractions.

Key Idea

Fractional distillation works because different hydrocarbons boil at different temperatures. Vapour rises up a tall column that is hottest at the bottom; each fraction condenses at the height where the temperature matches its boiling point.

Working Principle

Lighter fractions with low boiling points (LPG, petrol) are collected near the top; heavier fractions (diesel, lubricating oil, paraffin wax, bitumen) are collected lower down. Natural gas is mainly methane and is compressed to about 200 atmospheres to give CNG.

Applications

- Petroleum gas (LPG) — domestic and industrial fuel; petrol — motor fuel, dry cleaning solvent.
- Kerosene — stoves and jet fuel; diesel — heavy vehicles and generators; bitumen — road surfacing.
- CNG is used for power generation and as a clean vehicle fuel; it can be piped directly to homes.

Points to Remember (Exam Focus)

- ✓ Petroleum is called 'black gold' because of its enormous value.
- ✓ Refining = fractional distillation, separating fractions by their different boiling points.

- ✓ Lighter fractions (lower boiling point) are collected higher in the column.
- ✓ CNG is compressed natural gas — a cleaner fuel that causes far less pollution.
- ✓ Petrochemicals are substances obtained from petroleum and natural gas, e.g. detergents, plastics, fibres.

⚠ Common Student Mistakes

- ⚠ Thinking petroleum is a single compound — it is a mixture of many hydrocarbons.
- ⚠ Saying LPG and CNG are the same; LPG is mainly butane/propane, CNG is mainly methane.
- ⚠ Placing heavy fractions at the top of the fractionating column; they condense at the bottom.

🧠 Memory Hook

'Light rises High, Heavy stays Low' in the fractionating column — and 'PDS: Petrol Drives, Diesel Drags, Bitumen Beds the road'.

🚀 Advanced Insights

The Petroleum Conservation Research Association advises driving at a steady 40–60 km/h, keeping tyres correctly inflated and switching off at traffic lights. Each measure works by reducing the energy wasted as heat and drag rather than by changing the fuel itself.

II Quick Recap — Topics Covered So Far

- Petroleum, Natural Gas and Conservation — Petroleum is separated into useful fractions by fractional distillation in a refinery, natural gas is a clean fuel stored as CNG, and both must be conserved because they are running out.

Chapter 3 — End of Chapter Revision

All Memory Hooks

- Natural Resources and the Formation of Fossil Fuels: 'Coal from Trees, Petroleum from the Seas' — and coal grades 'Peat, Lignite, Bituminous, Anthracite' get better as you go.
- Coal and Its Products: 'Coal gives Coke, Coal Tar and Coal Gas — one Solid, one Liquid, one Gas.'
- Petroleum, Natural Gas and Conservation: 'Light rises High, Heavy stays Low' in the fractionating column — and 'PDS: Petrol Drives, Diesel Drags, Bitumen Beds the road'.

All Common Mistakes

-  Calling forests inexhaustible — they are exhaustible, though renewable if managed.
-  Saying coal is formed from dead animals; coal comes from plants, petroleum from marine organisms.
-  Confusing carbonisation (coal formation) with combustion (burning).
-  Saying destructive distillation happens in the presence of air — the absence of air is the whole point.
-  Confusing coke (from coal) with charcoal (from wood).
-  Thinking coal gas is the same as natural gas; coal gas is manufactured, natural gas is extracted.
-  Thinking petroleum is a single compound — it is a mixture of many hydrocarbons.
-  Saying LPG and CNG are the same; LPG is mainly butane/propane, CNG is mainly methane.
-  Placing heavy fractions at the top of the fractionating column; they condense at the bottom.

Quick Revision Section

- Natural Resources and the Formation of Fossil Fuels: Coal, petroleum and natural gas are fossil fuels formed over millions of years from buried remains of ancient plants and animals, and they are exhaustible natural resources.
- Coal and Its Products: Heating coal in the absence of air is called destructive distillation, and it yields coke, coal tar and coal gas — three products with distinct uses.
- Petroleum, Natural Gas and Conservation: Petroleum is separated into useful fractions by fractional distillation in a refinery, natural gas is a clean fuel stored as CNG, and both must be conserved because they are running out.

Chapter 4: Combustion and Flame

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge that oxygen is needed for burning.
- Idea that fuels release heat when burnt.
- Basic understanding of temperature and heat.
- Familiarity with a candle flame and a gas stove flame.

4.1 Combustion and Conditions Necessary for Burning

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Combustion is a chemical process in which a substance reacts with oxygen to give heat and usually light, and it needs three things — fuel, air and the ignition temperature.

Real-Life Example

A matchstick lights when friction supplies enough heat to reach its ignition temperature; blowing on a candle removes heat and the same flame dies.

Detailed Explanation

Definition

Combustion is the chemical process in which a substance reacts with oxygen to produce heat and light. The substance that undergoes combustion is called a combustible substance or fuel. The lowest temperature at which a substance catches fire is its ignition temperature.

Key Idea

All three conditions must be present together — fuel, a supporter of combustion (oxygen), and a temperature at or above the ignition temperature. Remove any one and combustion stops; every fire extinguisher works by removing one of the three.

Working Principle

Water extinguishes ordinary fires by cooling the fuel below its ignition temperature and by forming steam that cuts off air. Carbon dioxide extinguishers work differently — CO_2 is heavier than air, so it blankets the fire and cuts off oxygen, and it leaves no damaging residue.

Applications

- Water must NEVER be used on electrical or oil fires — it conducts electricity and oil floats on water, spreading the fire.
- CO_2 is the best extinguisher for electrical and oil fires.
- A fire in a forest may start from lightning or from the heat of dry, rubbing branches.

✓ Points to Remember (Exam Focus)

- ✓ Three requirements: fuel, oxygen (supporter of combustion), and ignition temperature.
- ✓ Ignition temperature is the LOWEST temperature at which a substance catches fire.
- ✓ Inflammable substances have very low ignition temperature: petrol, alcohol, LPG, spirit.
- ✓ Water extinguishes by cooling below the ignition temperature and cutting off air.
- ✓ CO₂ is used on electrical and oil fires because it does not conduct electricity and leaves no residue.

⚠ Common Student Mistakes

- ⚠ Saying 'a substance burns if you heat it enough' without mentioning oxygen — both are required.
- ⚠ Using water on an oil or electrical fire; it makes both worse.
- ⚠ Confusing ignition temperature (needed to start burning) with boiling point.

**Memory Hook**

'The Fire Triangle: Fuel + Oxygen + Heat' — break any side and the fire dies.

**Advanced Insights**

CO₂ extinguishers also cool as they discharge: the compressed gas expands rapidly and its temperature drops sharply, so a single extinguisher attacks two sides of the fire triangle — the oxygen supply and the heat — at once.

4.2 Types of Combustion and the Structure of a Flame

🕒 Estimated time: 14 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Combustion may be rapid, spontaneous or explosive, and a candle flame has three zones — a dark inner zone, a luminous yellow middle zone and a non-luminous blue outer zone which is the hottest.

**Real-Life Example**

Goldsmiths blow the outermost part of a flame onto gold to melt it, because the outer blue zone — not the bright yellow one — is the hottest part.

📖 Detailed Explanation

Definition

A flame is the region where the gaseous substances produced by a burning substance undergo combustion. Only substances that vaporise on heating burn with a flame.

Key Idea

The colour of a flame zone tells you how completely the fuel is burning. The outer blue zone has plenty of oxygen and burns completely, making it the hottest but non-luminous. The middle yellow zone has partial combustion producing glowing carbon particles, so it is bright but cooler.

Working Principle

Rapid combustion produces heat and light quickly (LPG burning). Spontaneous combustion occurs without any apparent cause (white phosphorus in air). Explosion produces a large amount of heat, light, sound and gas suddenly (firecrackers).

Applications

- Candle wax, kerosene and camphor burn with a flame; coal and charcoal glow without a flame.
- The unburnt carbon particles in the luminous zone deposit soot on a glass plate held in it.
- Blue flame of a gas stove = complete combustion; yellow flame = incomplete combustion and wasted fuel.

✓ Points to Remember (Exam Focus)

- ✓ Three zones of a candle flame: dark inner (unburnt vapour), luminous middle (partial combustion), non-luminous outer (complete combustion, hottest).
- ✓ The outermost blue zone is the HOTTEST part of the flame.
- ✓ Substances that vaporise on heating burn with a flame; those that do not, glow.
- ✓ Rapid combustion: LPG; Spontaneous: white phosphorus; Explosion: firecracker.
- ✓ A yellow, sooty flame indicates incomplete combustion and insufficient oxygen.

⚠ Common Student Mistakes

- ⚠ Saying the yellow luminous part is the hottest — it is the brightest but the outer blue zone is hottest.
- ⚠ Assuming every burning substance produces a flame; coal and charcoal glow without one.
- ⚠ Calling explosion the same as rapid combustion; explosion additionally produces sound and a large volume of gas.

**Memory Hook**

'Blue is Best (hottest), Yellow is Yeoman (bright but cool), Black is Blank (unburnt)' — outer, middle, inner.

**Advanced Insights**

The yellow of a candle flame is blackbody radiation from soot particles heated to about $1,400^{\circ}\text{C}$. A blue flame contains no soot, so its colour comes from electronic transitions in excited CH and C_2 molecules — different physics entirely, which is why colour is such a reliable indicator of combustion completeness.

II Quick Recap — Topics Covered So Far

- **Combustion and Conditions Necessary for Burning** — Combustion is a chemical process in which a substance reacts with oxygen to give heat and usually light, and it needs three things — fuel, air and the ignition temperature.
- **Types of Combustion and the Structure of a Flame** — Combustion may be rapid, spontaneous or explosive, and a candle flame has three zones — a dark inner zone, a luminous yellow middle zone and a non-luminous blue outer zone which is the hottest.

4.3 Fuels, Calorific Value and Fuel Efficiency

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A good fuel has a high calorific value, moderate ignition temperature, low cost, easy availability and produces little pollution; calorific value is the heat produced by burning 1 kg of a fuel.

Real-Life Example

Hydrogen releases nearly four times as much energy per kilogram as LPG, which is why rockets use it — but it is hard to store, so kitchens still use LPG.

Detailed Explanation

Definition

Calorific value is the amount of heat energy produced on completely burning 1 kilogram of a fuel. Its unit is kilojoule per kilogram (kJ/kg).

Key Idea

No single fuel is ideal for every job. The best fuel for a purpose is the one that balances calorific value, safety, cost, availability, storage and pollution — which is why aircraft, kitchens and power stations all use different fuels.

Working Principle

Incomplete combustion produces carbon monoxide, a poisonous gas that binds to haemoglobin and can kill in a closed room. It also wastes fuel, since not all the chemical energy is released.

Applications

- Burning fossil fuels releases CO_2 (global warming), SO_2 and NO_2 (acid rain), and unburnt carbon particles (respiratory disease).
- CNG and hydrogen are cleaner fuels; hydrogen has the highest calorific value of all fuels.

- Charcoal is preferred over wood because it has a higher calorific value and produces less smoke.

✓ Points to Remember (Exam Focus)

- ✓ Calorific value = heat produced by burning 1 kg of fuel completely, in kJ/kg.
- ✓ Hydrogen has the highest calorific value (about 150,000 kJ/kg).
- ✓ Carbon monoxide from incomplete combustion is poisonous and can be fatal in closed rooms.
- ✓ Burning coal and petroleum releases CO_2 , causing global warming.
- ✓ A good fuel: high calorific value, moderate ignition temperature, low ash, cheap, safe to store.

⚠ Common Student Mistakes

- ⚠ Saying a fuel with a very low ignition temperature is best — it would be dangerously easy to ignite.
- ⚠ Confusing carbon dioxide (greenhouse gas, not poisonous) with carbon monoxide (poisonous).
- ⚠ Thinking calorific value depends on the amount of fuel burnt; it is defined per kilogram.



Memory Hook

'High heat, Moderate ignition, Low cost, Low pollution' — the four marks of a good fuel.



Advanced Insights

Hydrogen's very high calorific value per kilogram hides a practical problem: per unit volume it stores far less energy than petrol, so it must be compressed or liquefied. Energy density by mass and by volume are different measures, and vehicles are constrained by both.

II Quick Recap — Topics Covered So Far

- Fuels, Calorific Value and Fuel Efficiency — A good fuel has a high calorific value, moderate ignition temperature, low cost, easy availability and produces little pollution; calorific value is the heat produced by burning 1 kg of a fuel.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Combustion and Conditions Necessary for Burning: 'The Fire Triangle: Fuel + Oxygen + Heat' — break any side and the fire dies.
- Types of Combustion and the Structure of a Flame: 'Blue is Best (hottest), Yellow is Yeoman (bright but cool), Black is Blank (unburnt)' — outer, middle, inner.
- Fuels, Calorific Value and Fuel Efficiency: 'High heat, Moderate ignition, Low cost, Low pollution' — the four marks of a good fuel.

All Common Mistakes

-  Saying 'a substance burns if you heat it enough' without mentioning oxygen — both are required.
-  Using water on an oil or electrical fire; it makes both worse.
-  Confusing ignition temperature (needed to start burning) with boiling point.
-  Saying the yellow luminous part is the hottest — it is the brightest but the outer blue zone is hottest.
-  Assuming every burning substance produces a flame; coal and charcoal glow without one.
-  Calling explosion the same as rapid combustion; explosion additionally produces sound and a large volume of gas.
-  Saying a fuel with a very low ignition temperature is best — it would be dangerously easy to ignite.
-  Confusing carbon dioxide (greenhouse gas, not poisonous) with carbon monoxide (poisonous).
-  Thinking calorific value depends on the amount of fuel burnt; it is defined per kilogram.

Quick Revision Section

- Combustion and Conditions Necessary for Burning: Combustion is a chemical process in which a substance reacts with oxygen to give heat and usually light, and it needs three things — fuel, air and the ignition temperature.
- Types of Combustion and the Structure of a Flame: Combustion may be rapid, spontaneous or explosive, and a candle flame has three zones — a dark inner zone, a luminous yellow middle zone and a non-luminous blue outer zone which is the hottest.
- Fuels, Calorific Value and Fuel Efficiency: A good fuel has a high calorific value, moderate ignition temperature, low cost, easy availability and produces little pollution; calorific value is the heat produced by burning 1 kg of a fuel.

Chapter 5: Language of Chemistry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that all matter is made of atoms.
- Knowledge that elements are the simplest pure substances.
- Basic arithmetic with whole numbers and ratios.
- Familiarity with the names of common elements such as oxygen and iron.

5.1 Symbols of Elements

 **Estimated time:** 12 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

A symbol is a short form of the name of an element — one or two letters, with the first always capital and the second always small.

Real-Life Example

The 'Fe' on an iron tablet and the 'Na' on a salt packet are chemical symbols in daily use, both taken from Latin rather than English names.

Detailed Explanation

Definition

A symbol is an abbreviated representation of the name of an element. It represents both the element and exactly one atom of it.

Key Idea

Berzelius introduced the modern system: the first letter of the English name, capitalised, and where two elements share that letter, a second small letter is added. Some symbols come from Latin names, and these are the ones most often forgotten.

Working Principle

A symbol carries quantitative meaning too — 'O' means one atom of oxygen with a mass of 16 u, while 'O₂' means one molecule of oxygen with a mass of 32 u.

Applications

- Latin-derived symbols: Na (Natrium, sodium), K (Kalium, potassium), Fe (Ferrum, iron), Cu (Cuprum, copper).
- More Latin symbols: Ag (Argentum, silver), Au (Aurum, gold), Pb (Plumbum, lead), Sn (Stannum, tin), Sb (Stibium, antimony), Hg (Hydrargyrum, mercury).
- Symbols are the same in every language, which makes chemistry universal.

✓ Points to Remember (Exam Focus)

- ✓ First letter of a symbol is ALWAYS capital; the second letter is ALWAYS small.
- ✓ A symbol represents one atom of the element and its atomic mass.
- ✓ Latin symbols to memorise: Na, K, Fe, Cu, Ag, Au, Pb, Sn, Hg, Sb, W.
- ✓ Berzelius introduced the modern system of symbols.
- ✓ Co is cobalt, but CO is carbon monoxide — capitalisation changes the meaning completely.

⚠ Common Student Mistakes

- ⚠ Writing CO for cobalt instead of Co — the capital O turns it into a compound.
- ⚠ Writing 'NA' or 'FE' — only the first letter is capital.
- ⚠ Guessing symbols from English names for Latin-derived elements (writing 'S' or 'So' for sodium).

**Memory Hook**

'Capital first, small second — always.' And for Latin ones: 'Sodium's Natrium, Potassium's Kalium, Iron's Ferrum, Copper's Cuprum'.

**Advanced Insights**

IUPAC now assigns three-letter placeholder symbols to newly synthesised superheavy elements until a permanent name is approved — for example Uut for element 113 before it became Nihonium (Nh). The naming convention itself is part of the language of chemistry.

5.2 Valency, Ions and Radicals

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Valency is the combining capacity of an element, an ion is a charged atom or group of atoms, and a radical is a group of atoms that behaves as a single unit in a reaction.

**Real-Life Example**

Sodium always combines one-to-one with chlorine to give NaCl, but calcium takes two chlorines to give CaCl_2 — valency is the reason.

**Detailed Explanation****Definition**

Valency is the number of electrons an atom loses, gains or shares to attain a stable configuration — that is, its combining capacity. An ion is an atom or group of atoms carrying a net charge; a positive ion is a cation and a negative ion is an anion.

Key Idea

A compound as a whole is electrically neutral, so the total positive charge must exactly balance the total negative charge. This single rule generates every chemical formula.

Working Principle

Radicals are groups of atoms that carry a charge and stay together through a reaction — sulphate (SO_4^{2-}), nitrate (NO_3^-), carbonate (CO_3^{2-}), ammonium (NH_4^+). They are treated as single units and bracketed when more than one is needed.

Applications

- Variable valency: iron shows +2 (ferrous, Fe^{2+}) and +3 (ferric, Fe^{3+}); copper shows +1 (cuprous) and +2 (cupric).
- The '-ous' suffix indicates the lower valency and '-ic' the higher valency.
- Ammonium (NH_4^+) is the only common positive radical.

✓ Points to Remember (Exam Focus)

- ✓ Valency = combining capacity; it has no sign, while charge on an ion does.
- ✓ Cation = positive (metal, loses electrons); Anion = negative (non-metal, gains electrons).
- ✓ Common monovalent radicals: OH^- , NO_3^- , HCO_3^- , NH_4^+ ; divalent: SO_4^{2-} , CO_3^{2-} ; trivalent: PO_4^{3-} .
- ✓ Noble gases have a valency of zero because their outermost shell is already complete.
- ✓ '-ous' = lower valency, '-ic' = higher valency (ferrous Fe^{2+} , ferric Fe^{3+}).

⚠ Common Student Mistakes

- ⚠ Writing brackets around a single radical — $\text{Na}(\text{NO}_3)$ should simply be NaNO_3 .
- ⚠ Forgetting brackets when a radical is taken more than once — writing CaNO_{32} instead of $\text{Ca}(\text{NO}_3)_2$.
- ⚠ Confusing valency (a number) with charge (a number with a sign).



Memory Hook

'CATion is PAWSitive (cat has paws), ANion is Negative' — and 'ous is LOWer, ic is higher'.



Advanced Insights

Valency is a bookkeeping idea; oxidation number is its more general successor and can be negative or fractional. In Fe_3O_4 , for instance, the average oxidation number of iron is +8/3 because the compound is really a mixture of Fe^{2+} and Fe^{3+} ions.

II Quick Recap — Topics Covered So Far

- Symbols of Elements — A symbol is a short form of the name of an element — one or two letters, with the first always capital and the second always small.
- Valency, Ions and Radicals — Valency is the combining capacity of an element, an ion is a charged atom or group of atoms, and a radical is a group of atoms that behaves as a single unit in a reaction.

5.3 Chemical Formulae and Chemical Equations

Estimated time: 16 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

A chemical formula shows the elements in a compound and how many atoms of each are present, and it is written by the criss-cross method of exchanging valencies.

Real-Life Example

Writing H_2SO_4 instead of 'sulphuric acid' tells a chemist anywhere in the world exactly which atoms are present and in what ratio.

Detailed Explanation

Definition

A chemical formula is the symbolic representation of the composition of a molecule of a substance, showing which elements are present and in what ratio.

Key Idea

In the criss-cross method the valency of each element or radical is written above it, and then the two valencies are exchanged and written as subscripts. The subscripts are then reduced to the simplest whole-number ratio.

Working Principle

The positive part (metal or positive radical) is always written first, followed by the negative part. If a radical has to be taken more than once, it is enclosed in brackets.

Applications

- Al (valency 3) and SO_4 (valency 2) give $\text{Al}_2(\text{SO}_4)_3$.
- Ca (valency 2) and O (valency 2) give CaO after reducing Ca_2O_2 to its simplest ratio.
- Acids: HCl, H_2SO_4 , HNO_3 , H_2CO_3 ; Bases: NaOH, KOH, $\text{Ca}(\text{OH})_2$, NH_4OH .

✓ Points to Remember (Exam Focus)

- ✓ Write the positive part first, negative part second, then criss-cross the valencies.
- ✓ Always reduce the subscripts to the simplest whole-number ratio.
- ✓ Use brackets when a radical is taken more than once: $\text{Ca}(\text{OH})_2$, $\text{Al}_2(\text{SO}_4)_3$.

- ✓ A chemical equation must be balanced — the same number of atoms of each element on both sides.
- ✓ Only coefficients may be changed while balancing, never the subscripts in a formula.

⚠ Common Student Mistakes

- ⚠ Changing a subscript while balancing an equation, which changes the substance itself.
- ⚠ Forgetting to reduce Ca_2O_2 to CaO after criss-crossing equal valencies.
- ⚠ Writing the negative radical first — the metal or positive radical always comes first.



Memory Hook

'Criss-cross the valencies, then cancel down' — and 'Balance by coefficients, never by subscripts'.



Advanced Insights

Balancing an equation is really an application of the law of conservation of mass. For complex redox equations the ion-electron method is used instead of trial and error — balancing charge and atoms separately, which is far more systematic.

II Quick Recap — Topics Covered So Far

- Chemical Formulae and Chemical Equations — A chemical formula shows the elements in a compound and how many atoms of each are present, and it is written by the criss-cross method of exchanging valencies.

Chapter 5 — End of Chapter Revision

All Memory Hooks

- Symbols of Elements: 'Capital first, small second — always.' And for Latin ones: 'Sodium's Natrium, Potassium's Kalium, Iron's Ferrum, Copper's Cuprum'.
- Valency, Ions and Radicals: 'CATion is PAWSitive (cat has paws), ANion is Negative' — and 'ous is IOWer, ic is higher'.
- Chemical Formulae and Chemical Equations: 'Criss-cross the valencies, then cancel down' — and 'Balance by coefficients, never by subscripts'.

All Common Mistakes

-  Writing CO for cobalt instead of Co — the capital O turns it into a compound.
-  Writing 'NA' or 'FE' — only the first letter is capital.
-  Guessing symbols from English names for Latin-derived elements (writing 'S' or 'So' for sodium).
-  Writing brackets around a single radical — $\text{Na}(\text{NO}_3)$ should simply be NaNO_3 .
-  Forgetting brackets when a radical is taken more than once — writing CaNO_{32} instead of $\text{Ca}(\text{NO}_3)_2$.
-  Confusing valency (a number) with charge (a number with a sign).
-  Changing a subscript while balancing an equation, which changes the substance itself.
-  Forgetting to reduce Ca_2O_2 to CaO after criss-crossing equal valencies.
-  Writing the negative radical first — the metal or positive radical always comes first.

Quick Revision Section

- Symbols of Elements: A symbol is a short form of the name of an element — one or two letters, with the first always capital and the second always small.
- Valency, Ions and Radicals: Valency is the combining capacity of an element, an ion is a charged atom or group of atoms, and a radical is a group of atoms that behaves as a single unit in a reaction.
- Chemical Formulae and Chemical Equations: A chemical formula shows the elements in a compound and how many atoms of each are present, and it is written by the criss-cross method of exchanging valencies.

Chapter 6: Classification of Matter on the Basis of Chemical Composition

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that matter is anything that has mass and occupies space.
- Knowledge of chemical symbols and formulae.
- Familiarity with common substances such as water, salt and air.
- Basic understanding that some substances can be separated and others cannot.

6.1 Pure Substances — Elements and Compounds

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

A pure substance has a fixed composition throughout and is either an element, which cannot be broken down further, or a compound, formed by chemically combining elements in a fixed ratio.

Real-Life Example

Pure gold is an element, table salt is a compound, and sea water is neither — it is a mixture, even though all three look uniform.

Detailed Explanation

Definition

An element is a pure substance made of only one kind of atom and cannot be broken into simpler substances by chemical means. A compound is a pure substance formed when two or more elements combine chemically in a fixed ratio by mass.

Key Idea

In a compound the constituents lose their individual properties completely. Sodium is a metal that catches fire in water and chlorine is a poisonous gas, yet sodium chloride is a harmless substance we eat every day.

Working Principle

Elements are classified as metals, non-metals and metalloids. Atomicity is the number of atoms in one molecule of an element: monoatomic (He, Ne, Ar), diatomic (H_2 , O_2 , N_2 , Cl_2), triatomic (O_3), tetraatomic (P_4), polyatomic (S_8).

Applications

- There are 118 known elements, of which about 94 occur naturally.
- Compounds are formed with the absorption or evolution of energy — a chemical change.
- The composition of a compound is always fixed: water is always 1:8 hydrogen to oxygen by mass.

✓ Points to Remember (Exam Focus)

- ✓ Element = one kind of atom; Compound = two or more elements chemically combined in a fixed ratio.
- ✓ In a compound, the constituents lose their original properties.
- ✓ A compound has a fixed melting point and boiling point; a mixture does not.
- ✓ Atomicity: He (mono), O₂ (di), O₃ (tri), P₄ (tetra), S₈ (poly).
- ✓ Compounds can be separated only by chemical methods, never by physical ones.

⚠ Common Student Mistakes

- ⚠ Calling air a compound — it is a mixture, since its composition varies and its gases keep their properties.
- ⚠ Saying a compound is a mixture of elements; the elements are chemically bonded, not merely mixed.
- ⚠ Thinking all elements are monoatomic — most non-metals exist as diatomic or polyatomic molecules.

**Memory Hook**

'Element = One; Compound = Combined and Constant ratio; Mixture = Mixed and Mutable.'

**Advanced Insights**

The law of definite proportions — that a compound always contains the same elements in the same mass ratio — was the crucial evidence for Dalton's atomic theory. If matter were continuous rather than made of discrete atoms, there would be no reason for such fixed ratios.

6.2 Mixtures — Homogeneous and Heterogeneous

⌚ Estimated time: 13 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

A mixture contains two or more substances in any proportion that keep their own properties; it is homogeneous if uniform throughout and heterogeneous if not.

**Real-Life Example**

Sugar dissolved in tea is homogeneous — every sip tastes the same; a bowl of mixed dry fruits is heterogeneous — every handful is different.

📖 Detailed Explanation

Definition

A mixture is a substance containing two or more pure substances mixed in any proportion, which can be separated by physical methods. A homogeneous mixture has a uniform composition throughout; a heterogeneous mixture does not.

Key Idea

The key contrast with a compound is threefold: a mixture has variable composition, its constituents keep their own properties, and it can be separated by physical means without any chemical reaction.

Working Principle

A solution is a homogeneous mixture of a solute dissolved in a solvent. Air, brass, salt water and soda water are homogeneous; sand and water, oil and water, and soil are heterogeneous.

Applications

- Alloys such as brass and steel are homogeneous mixtures — solid solutions.
- Air is a homogeneous mixture of nitrogen (78%), oxygen (21%) and other gases.
- Milk is a colloid — it looks homogeneous but is heterogeneous at the microscopic level.

✓ Points to Remember (Exam Focus)

- ✓ Mixture: variable composition, constituents keep properties, separable by physical means.
- ✓ Homogeneous = uniform throughout (solutions, alloys, air).
- ✓ Heterogeneous = non-uniform (sand + water, oil + water, soil).
- ✓ No energy is absorbed or released in forming a mixture.
- ✓ A mixture has no fixed melting or boiling point.

⚠ Common Student Mistakes

- ⚠ Calling every clear liquid a solution; some clear liquids are pure substances.
- ⚠ Assuming alloys are compounds — they are homogeneous mixtures.
- ⚠ Saying mixtures always look non-uniform; homogeneous mixtures look completely uniform.

**Memory Hook**

'HOMOgeneous = HOMOgenised milk look — one phase; HETEROgeneous = you can see the parts.'

**Advanced Insights**

Whether a mixture counts as homogeneous depends on the scale of observation. Milk appears uniform to the eye but a microscope reveals suspended fat globules — which is why colloids sit awkwardly between the two categories and are usually treated separately.

- **Pure Substances — Elements and Compounds** — A pure substance has a fixed composition throughout and is either an element, which cannot be broken down further, or a compound, formed by chemically combining elements in a fixed ratio.
- **Mixtures — Homogeneous and Heterogeneous** — A mixture contains two or more substances in any proportion that keep their own properties; it is homogeneous if uniform throughout and heterogeneous if not.

6.3 Separating the Components of a Mixture

🕒 **Estimated time:** 16 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Mixtures are separated by physical methods chosen according to the difference in properties — size, density, solubility, boiling point, magnetism or sublimation.

Real-Life Example

Winnowing grain in a village, filtering tea leaves through a strainer, and boiling sea water to get salt are three separation techniques used every day.

Detailed Explanation

Definition

Separation techniques are physical methods that exploit a difference in a physical property between the components of a mixture.

Key Idea

The method is always chosen by asking which single property differs most. Different densities suggest sedimentation or winnowing; different solubilities suggest filtration or crystallisation; different boiling points suggest distillation.

Working Principle

Filtration separates an insoluble solid from a liquid. Evaporation recovers a dissolved solid. Distillation recovers the solvent as well, by boiling and then condensing it. Fractional distillation separates two miscible liquids whose boiling points differ. Chromatography separates dissolved substances that travel at different rates on paper.

Applications

- Sublimation separates camphor, naphthalene or ammonium chloride from common salt.
- A magnet separates iron filings from sulphur powder.
- Chromatography separates the dyes in ink and the pigments in a leaf.

✓ Points to Remember (Exam Focus)

- ✓ Filtration: insoluble solid from liquid (chalk and water).

- ✓ Evaporation: soluble solid from solution (salt from sea water).
- ✓ Distillation: pure solvent recovered (pure water from salt water).
- ✓ Fractional distillation: two miscible liquids with different boiling points (petroleum, liquid air).
- ✓ Sublimation: a subliming solid from a non-subliming one (camphor from salt).

Common Student Mistakes

-  Using filtration to separate a dissolved solid — a solute passes straight through the filter paper.
-  Confusing evaporation (solvent lost) with distillation (solvent recovered).
-  Thinking centrifugation and filtration are interchangeable; centrifugation is for very fine suspended particles.



Memory Hook

'Match the Method to the Mismatch' — find the one property that differs and pick the technique that exploits it.



Advanced Insights

Chromatography works because each component has its own balance between sticking to the stationary phase and dissolving in the moving one. The same principle, scaled up, underlies HPLC and gas chromatography — the standard tools of every modern analytical laboratory.

II Quick Recap — Topics Covered So Far

- Separating the Components of a Mixture — Mixtures are separated by physical methods chosen according to the difference in properties — size, density, solubility, boiling point, magnetism or sublimation.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Pure Substances — Elements and Compounds: 'Element = One; Compound = Combined and Constant ratio; Mixture = Mixed and Mutable.'
- Mixtures — Homogeneous and Heterogeneous: 'HOMOgeneous = HOMOgenised milk look — one phase; HETEROgeneous = you can see the parts.'
- Separating the Components of a Mixture: 'Match the Method to the Mismatch' — find the one property that differs and pick the technique that exploits it.

All Common Mistakes

-  Calling air a compound — it is a mixture, since its composition varies and its gases keep their properties.
-  Saying a compound is a mixture of elements; the elements are chemically bonded, not merely mixed.
-  Thinking all elements are monoatomic — most non-metals exist as diatomic or polyatomic molecules.
-  Calling every clear liquid a solution; some clear liquids are pure substances.
-  Assuming alloys are compounds — they are homogeneous mixtures.
-  Saying mixtures always look non-uniform; homogeneous mixtures look completely uniform.
-  Using filtration to separate a dissolved solid — a solute passes straight through the filter paper.
-  Confusing evaporation (solvent lost) with distillation (solvent recovered).
-  Thinking centrifugation and filtration are interchangeable; centrifugation is for very fine suspended particles.

Quick Revision Section

- Pure Substances — Elements and Compounds: A pure substance has a fixed composition throughout and is either an element, which cannot be broken down further, or a compound, formed by chemically combining elements in a fixed ratio.
- Mixtures — Homogeneous and Heterogeneous: A mixture contains two or more substances in any proportion that keep their own properties; it is homogeneous if uniform throughout and heterogeneous if not.
- Separating the Components of a Mixture: Mixtures are separated by physical methods chosen according to the difference in properties — size, density, solubility, boiling point, magnetism or sublimation.

Chapter 7: States of Matter

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that matter is made of tiny particles.
- Familiarity with ice, water and steam as three forms of the same substance.
- Knowledge that heating raises temperature.
- Basic understanding of the words melting and boiling.

7.1 Particle Nature and the Three States of Matter

 **Estimated time:** 15 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Matter is made of tiny particles with spaces between them that are constantly moving, and the three states — solid, liquid and gas — differ in how strongly those particles are held together.

Real-Life Example

The smell of incense reaching across a room, and sugar disappearing into a glass of water without raising the level much, both show that particles move and that there are gaps between them.

Detailed Explanation

Definition

Matter is anything that has mass and occupies space. Diffusion is the spontaneous intermixing of the particles of two substances; it proves that particles are in constant motion and have spaces between them.

Key Idea

Five postulates of the particle theory: matter is made of very small particles; there are spaces between them; they are in constant motion; they attract one another; and their kinetic energy increases with temperature.

Working Principle

In solids the intermolecular force is strongest and the spaces smallest, so particles only vibrate about fixed positions. In liquids the force is weaker, so particles slide past one another. In gases the force is negligible, so particles move freely and fill any container.

Applications

- Diffusion is fastest in gases, slower in liquids, and negligible in solids.
- Gases are highly compressible, which is why LPG and oxygen are supplied in cylinders.
- Solids have a definite shape and volume; liquids have definite volume only; gases have neither.

✓ Points to Remember (Exam Focus)

- ✓ Order of intermolecular force: Solid > Liquid > Gas.
- ✓ Order of intermolecular space and kinetic energy: Gas > Liquid > Solid.
- ✓ Solids: definite shape and volume, incompressible, high density.
- ✓ Liquids: definite volume, no definite shape, take the shape of the container.
- ✓ Gases: neither definite shape nor volume, highly compressible, low density.

⚠ Common Student Mistakes

- ⚠ Saying liquids cannot be compressed at all — they are only very slightly compressible.
- ⚠ Thinking particles in a solid are completely stationary; they vibrate about fixed positions.
- ⚠ Confusing diffusion (mixing due to particle motion) with dissolution (forming a solution).

**Memory Hook**

'Solid = Strongly held, Liquid = Loosely held, Gas = Get away free.'

**Advanced Insights**

Plasma and Bose–Einstein condensate are the fourth and fifth states of matter. Plasma — ionised gas — is actually the most common state in the universe, making up stars and the glow inside fluorescent tubes and neon signs.

7.2 Interconversion of States — Melting, Boiling and Latent Heat

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Matter changes state when temperature or pressure changes, and the heat absorbed during a change of state without any rise in temperature is called latent heat.

**Real-Life Example**

Ice at 0 °C keeps a drink far colder than water at 0 °C, because melting ice absorbs a large amount of latent heat from the drink without itself getting warmer.

**Detailed Explanation****Definition**

Melting point is the temperature at which a solid becomes a liquid; boiling point is the temperature at which a liquid becomes a gas throughout its bulk. Latent heat is the heat energy absorbed or released during a change of state at constant temperature.

Key Idea

During melting or boiling the temperature stays constant even though heat is still being supplied. That heat is not raising kinetic energy — it is being used to overcome the forces of attraction between particles.

Working Principle

Latent heat of fusion of ice is 3.34×10^5 J/kg; latent heat of vaporisation of water is 22.5×10^5 J/kg. Vaporisation needs far more energy because the particles must be separated completely, not merely loosened.

Applications

- Steam at 100 °C causes far more severe burns than water at 100 °C because of its latent heat.
- Sublimation is the direct change from solid to gas — camphor, naphthalene, dry ice, iodine, ammonium chloride.
- Increasing pressure and lowering temperature liquefies a gas; this is how LPG is stored.

✓ Points to Remember (Exam Focus)

- ✓ Melting point of ice = 0 °C = 273 K; boiling point of water = 100 °C = 373 K.
- ✓ Kelvin conversion: $K = ^\circ C + 273$; $^\circ C = K - 273$.
- ✓ Temperature stays constant during a change of state, even though heat is supplied.
- ✓ Latent heat of vaporisation is much larger than latent heat of fusion.
- ✓ Sublimation: solid → gas directly, without becoming liquid.

⚠ Common Student Mistakes

- ⚠ Expecting the temperature to keep rising while ice is melting — it stays at 0 °C until all the ice has melted.
- ⚠ Subtracting 273 when converting Celsius to Kelvin instead of adding it.
- ⚠ Confusing evaporation (surface, any temperature) with boiling (bulk, at the boiling point).

Memory Hook

'Latent = Hidden heat — it hides in the change of state, not in the thermometer.'

Advanced Insights

Water's exceptionally high latent heat of vaporisation comes from hydrogen bonding. It is why sweating cools the body so effectively, why coastal climates are mild, and why steam carries so much energy in a power station.

Formula Section

Celsius–Kelvin Conversion

Difficulty	● Easy ★☆☆
Formula	$K = ^\circ C + 273$
Meaning of Variables	K = temperature on the Kelvin scale, $^\circ C$ = temperature on the Celsius scale.
When to Use	Whenever a temperature has to be expressed on the absolute (SI) scale, or converted back.
Common Mistakes	Subtracting instead of adding, and writing a degree sign with K — it is written 300 K, not 300 $^\circ K$.
Shortcut / Memory Trick	Kelvin is always the bigger number for everyday temperatures — add 273 going up to K.

Heat Absorbed During Change of State

Difficulty	● Medium ★★★
Formula	$Q = m \times L$
Meaning of Variables	Q = heat energy in joules, m = mass in kg, L = specific latent heat in J/kg.
When to Use	To find the heat needed to melt or vaporise a given mass, with no change in temperature.
Common Mistakes	Using the latent heat of fusion where vaporisation is meant, or including a temperature term — there is no ΔT in this formula.
Shortcut / Memory Trick	No temperature change $\rightarrow$ use $Q = mL$; temperature change $\rightarrow$ use $Q = mc\Delta T$.

II Quick Recap — Topics Covered So Far

- Particle Nature and the Three States of Matter — Matter is made of tiny particles with spaces between them that are constantly moving, and the three states — solid, liquid and gas — differ in how strongly those particles are held together.
- Interconversion of States — Melting, Boiling and Latent Heat — Matter changes state when temperature or pressure changes, and the heat absorbed during a change of state without any rise in temperature is called latent heat.

7.3 Evaporation and Its Cooling Effect

 Estimated time: 12 mins **Difficulty:** ● Medium ★★★ **Exam Importance:** ★★★★★

Quick Summary

Evaporation is the change of a liquid into vapour at the surface at any temperature below the boiling point, and it produces cooling because the fastest particles escape.



Real-Life Example

Water stays cool in an earthen pot, wet clothes dry faster on a windy day, and a splash of spirit on the palm feels icy — all evaporation.



Detailed Explanation

Definition

Evaporation is the slow change of a liquid into its vapour at the surface, occurring at all temperatures below the boiling point.

Key Idea

Only the particles at the surface with the highest kinetic energy escape. The particles left behind therefore have a lower average kinetic energy, and lower average kinetic energy means a lower temperature — so the liquid cools.

Working Principle

Evaporation increases with a larger surface area, a higher temperature, a lower humidity and a faster wind speed. Humidity slows it because air already saturated with vapour cannot accept more.

Applications

- An earthen pot keeps water cool because water seeping through its pores evaporates.
- We sweat in summer; evaporation of sweat cools the body.
- Clothes are spread out to dry to increase the exposed surface area.

✓ Points to Remember (Exam Focus)

- ✓ Evaporation is a surface phenomenon; boiling is a bulk phenomenon.
- ✓ Evaporation occurs at ALL temperatures; boiling occurs only at the boiling point.
- ✓ Factors that increase evaporation: larger surface area, higher temperature, lower humidity, more wind.
- ✓ Evaporation causes cooling because the most energetic particles leave.
- ✓ High humidity slows evaporation, which is why sweat does not dry in the monsoon.

⚠ Common Student Mistakes

- ⚠ Saying evaporation only happens at the boiling point — that is boiling, not evaporation.
- ⚠ Thinking humidity speeds up drying; it slows it down.
- ⚠ Explaining the cooling by 'water is cold' rather than by the loss of high-energy particles.

 Memory Hook

'The fastest runners escape, so the crowd left behind is slower — and slower means cooler.'

 Advanced Insights

Evaporative cooling sets a physical limit called the wet-bulb temperature. When humidity is so high that sweat cannot evaporate, the body cannot shed heat at all — which is why a wet-bulb temperature above about 35 °C is survivable only for a few hours.

II Quick Recap — Topics Covered So Far

- **Evaporation and Its Cooling Effect** — Evaporation is the change of a liquid into vapour at the surface at any temperature below the boiling point, and it produces cooling because the fastest particles escape.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Particle Nature and the Three States of Matter: 'Solid = Strongly held, Liquid = Loosely held, Gas = Get away free.'
- Interconversion of States — Melting, Boiling and Latent Heat: 'Latent = Hidden heat — it hides in the change of state, not in the thermometer.'
- Evaporation and Its Cooling Effect: 'The fastest runners escape, so the crowd left behind is slower — and slower means cooler.'

All Common Mistakes

-  Saying liquids cannot be compressed at all — they are only very slightly compressible.
-  Thinking particles in a solid are completely stationary; they vibrate about fixed positions.
-  Confusing diffusion (mixing due to particle motion) with dissolution (forming a solution).
-  Expecting the temperature to keep rising while ice is melting — it stays at 0 °C until all the ice has melted.
-  Subtracting 273 when converting Celsius to Kelvin instead of adding it.
-  Confusing evaporation (surface, any temperature) with boiling (bulk, at the boiling point).
-  Saying evaporation only happens at the boiling point — that is boiling, not evaporation.
-  Thinking humidity speeds up drying; it slows it down.
-  Explaining the cooling by 'water is cold' rather than by the loss of high-energy particles.

Formula Sheet

Formula	Meaning / Use
Celsius–Kelvin Conversion: $K = ^\circ C + 273$	Whenever a temperature has to be expressed on the absolute (SI) scale, or converted back.
Heat Absorbed During Change of State: $Q = m \times L$	To find the heat needed to melt or vaporise a given mass, with no change in temperature.

Quick Revision Section

- Particle Nature and the Three States of Matter: Matter is made of tiny particles with spaces between them that are constantly moving, and the three states — solid, liquid and gas — differ in how strongly those particles are held together.

- Interconversion of States — Melting, Boiling and Latent Heat: Matter changes state when temperature or pressure changes, and the heat absorbed during a change of state without any rise in temperature is called latent heat.
- Evaporation and Its Cooling Effect: Evaporation is the change of a liquid into vapour at the surface at any temperature below the boiling point, and it produces cooling because the fastest particles escape.

Chapter 8: Transformation of Matter

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge of chemical symbols, formulae and valency.
- Idea that matter exists in three states.
- Familiarity with rusting, burning and the souring of milk.
- Basic understanding of a chemical equation.

8.1 Physical and Chemical Changes

 **Estimated time:** 13 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

A physical change alters only the form or state of a substance and is usually reversible, while a chemical change produces one or more entirely new substances and is generally irreversible.

Real-Life Example

Melting wax is a physical change — it can be re-solidified; burning the same wax is a chemical change — the flame's products can never be turned back into a candle.

Detailed Explanation

Definition

A physical change is a change in which no new substance is formed and the composition remains the same. A chemical change, or chemical reaction, is one in which one or more new substances with new properties are formed.

Key Idea

The decisive test is whether a new substance has been formed. Colour change, evolution of gas, formation of a precipitate, change in temperature and change in smell are the usual signs of a chemical change.

Working Principle

In a physical change only the arrangement of molecules alters; the molecules themselves are unchanged. In a chemical change bonds are broken and new bonds formed, so the molecules themselves are different afterwards.

Applications

- Physical: melting of ice, boiling of water, dissolving sugar, breaking glass, magnetising iron.
- Chemical: rusting of iron, burning of paper, cooking food, digestion, souring of milk, photosynthesis.
- Both together: burning a candle — the wax melts (physical) and then burns (chemical).

✓ Points to Remember (Exam Focus)

- ✓ Physical change: no new substance, usually reversible, mass unchanged.
- ✓ Chemical change: new substance formed, usually irreversible, energy absorbed or released.
- ✓ Signs of a chemical change: gas evolved, precipitate formed, colour changed, heat or light produced.
- ✓ Dissolving salt in water is physical — the salt can be recovered by evaporation.
- ✓ Burning of a candle involves both a physical and a chemical change.

⚠ Common Student Mistakes

- ⚠ Calling every irreversible change chemical — breaking a glass is irreversible but physical.
- ⚠ Thinking a colour change always means a chemical change; some physical changes also alter appearance.
- ⚠ Saying dissolving sugar is chemical because it 'disappears' — it can be recovered unchanged.

**Memory Hook**

'New substance = chemical; New form = physical' — ask 'what is it now?' not 'can I undo it?'.

**Advanced Insights**

The boundary is not always sharp. Dissolving common salt in water is called physical, yet it does break the ionic lattice into separate hydrated ions. Chemists therefore judge by whether the original substance can be recovered unchanged by physical means.

8.2 Types of Chemical Reactions

🕒 **Estimated time:** 16 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Chemical reactions are classified as combination, decomposition, displacement, double displacement and redox, according to how atoms are rearranged.

**Real-Life Example**

The white light of a burning magnesium ribbon (combination), the fizz when baking soda meets vinegar (double displacement), and the copper coating on an iron nail in blue solution (displacement) are three types in three seconds.

**Detailed Explanation**

Definition

Combination: two or more substances combine to form a single product. Decomposition: a single compound breaks into two or more simpler substances. Displacement: a more reactive element displaces a less reactive one from its compound. Double displacement: two compounds exchange their ions.

Key Idea

Reactions are also classified by energy: exothermic reactions release heat (burning, respiration, slaking of lime) and endothermic reactions absorb heat (decomposition of calcium carbonate, photosynthesis).

Working Principle

Decomposition can be brought about by heat (thermal), electricity (electrolytic) or light (photolytic). Double displacement reactions in which an insoluble solid forms are called precipitation reactions; those between an acid and a base are neutralisation reactions.

Applications

- Combination: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$, with a dazzling white light.
- Decomposition: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ on heating; $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ by electrolysis.
- Double displacement: $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4\downarrow + 2\text{NaCl}$, a white precipitate.

✓ Points to Remember (Exam Focus)

- ✓ Combination is the reverse of decomposition.
- ✓ Displacement involves one element and one compound; double displacement involves two compounds.
- ✓ A precipitation reaction forms an insoluble solid, shown with a downward arrow ($\downarrow$).
- ✓ Exothermic = heat released; Endothermic = heat absorbed.
- ✓ Respiration is exothermic; photosynthesis is endothermic.

⚠ Common Student Mistakes

- ⚠ Calling every reaction that produces heat a combination reaction; the two classifications are independent.
- ⚠ Confusing displacement (one element replaced) with double displacement (ions exchanged).
- ⚠ Forgetting to balance the equation after identifying the reaction type.

**Memory Hook**

'Combine = come together; Decompose = come apart; Displace = push one out; Double displace = swap partners.'

**Advanced Insights**

Almost every reaction type can be re-described in terms of electron transfer. Combination, decomposition and displacement reactions are frequently redox reactions in disguise, whereas double displacement reactions usually are not — no element changes its oxidation number.

II Quick Recap — Topics Covered So Far

- **Physical and Chemical Changes** — A physical change alters only the form or state of a substance and is usually reversible, while a chemical change produces one or more entirely new substances and is generally irreversible.
- **Types of Chemical Reactions** — Chemical reactions are classified as combination, decomposition, displacement, double displacement and redox, according to how atoms are rearranged.

8.3 Oxidation, Reduction and Redox Reactions

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Oxidation is the addition of oxygen or removal of hydrogen, reduction is the reverse, and since the two always occur together the combined process is called a redox reaction.

Real-Life Example

A cut apple turning brown, butter going rancid and iron rusting are all everyday oxidation — which is why antioxidants are added to packaged food.

Detailed Explanation

Definition

Oxidation is the addition of oxygen or removal of hydrogen from a substance (or loss of electrons). Reduction is the addition of hydrogen or removal of oxygen (or gain of electrons). A reaction in which both occur simultaneously is a redox reaction.

Key Idea

Oxidation and reduction can never occur alone. The substance that gives oxygen (or takes electrons) is the oxidising agent and is itself reduced; the substance that removes oxygen is the reducing agent and is itself oxidised.

Working Principle

In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, copper oxide loses oxygen (reduced) and hydrogen gains it (oxidised). So CuO is the oxidising agent and H_2 is the reducing agent — each does the opposite of what happens to it.

Applications

- Corrosion of metals and rancidity of oily food are both oxidation processes.
- Rancidity is prevented by adding antioxidants, packing in nitrogen gas, and refrigeration.
- Extraction of metals uses carbon or carbon monoxide as a reducing agent.

✓ Points to Remember (Exam Focus)

- ✓ Oxidation = gain of oxygen / loss of hydrogen / loss of electrons.
- ✓ Reduction = loss of oxygen / gain of hydrogen / gain of electrons.
- ✓ The oxidising agent is itself REDUCED; the reducing agent is itself OXIDISED.
- ✓ Common oxidising agents: O_2 , $KMnO_4$, H_2O_2 ; reducing agents: H_2 , C, CO.
- ✓ Rancidity is the oxidation of fats and oils, giving a bad smell and taste.

⚠ Common Student Mistakes

- ⚠ Saying the oxidising agent gets oxidised — it gets reduced; that is the whole point of the name.
- ⚠ Treating oxidation and reduction as separate reactions; they always occur together.
- ⚠ Thinking oxidation always requires oxygen; loss of electrons counts even where no oxygen is present.

**Memory Hook**

'OIL RIG' — Oxidation Is Loss of electrons, Reduction Is Gain — and 'the agent does the opposite to itself'.

**Advanced Insights**

Packets of chips are filled with nitrogen rather than air precisely to exclude oxygen and delay rancidity. Nitrogen is chosen because it is chemically inert, plentiful and cheap — the same reasoning behind nitrogen-filled aircraft tyres and museum display cases.

II Quick Recap — Topics Covered So Far

- Oxidation, Reduction and Redox Reactions — Oxidation is the addition of oxygen or removal of hydrogen, reduction is the reverse, and since the two always occur together the combined process is called a redox reaction.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- Physical and Chemical Changes: 'New substance = chemical; New form = physical' — ask 'what is it now?' not 'can I undo it?'.
- Types of Chemical Reactions: 'Combine = come together; Decompose = come apart; Displace = push one out; Double displace = swap partners.'
- Oxidation, Reduction and Redox Reactions: 'OIL RIG' — Oxidation Is Loss of electrons, Reduction Is Gain — and 'the agent does the opposite to itself'.

All Common Mistakes

-  Calling every irreversible change chemical — breaking a glass is irreversible but physical.
-  Thinking a colour change always means a chemical change; some physical changes also alter appearance.
-  Saying dissolving sugar is chemical because it 'disappears' — it can be recovered unchanged.
-  Calling every reaction that produces heat a combination reaction; the two classifications are independent.
-  Confusing displacement (one element replaced) with double displacement (ions exchanged).
-  Forgetting to balance the equation after identifying the reaction type.
-  Saying the oxidising agent gets oxidised — it gets reduced; that is the whole point of the name.
-  Treating oxidation and reduction as separate reactions; they always occur together.
-  Thinking oxidation always requires oxygen; loss of electrons counts even where no oxygen is present.

Quick Revision Section

- Physical and Chemical Changes: A physical change alters only the form or state of a substance and is usually reversible, while a chemical change produces one or more entirely new substances and is generally irreversible.
- Types of Chemical Reactions: Chemical reactions are classified as combination, decomposition, displacement, double displacement and redox, according to how atoms are rearranged.
- Oxidation, Reduction and Redox Reactions: Oxidation is the addition of oxygen or removal of hydrogen, reduction is the reverse, and since the two always occur together the combined process is called a redox reaction.